

Lecturer:

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Wed., 15. May 2024

Deadline

Sun., 26. May 2024 (23:55)

Upload to ADAM

Modalities

The exercises have to be done in **groups of 2 students**. Upload only the final version of the exercise to Adam and specify your group partner.

Please upload your implementations for suggested API, requested scripts and your short report, containing all answers to the asked questions as a .pdf file to ADAM by the deadline at the latest.

Note on Screenshots:

Random screenshots without proper explanation will not be accepted as a solution. Please crop your images so they display only the relevant content or use highlighting or additional text to bring everything into context in your report. Ensure that text in images is readable and not too small or blurry.

Task 1 - SSH key-based authentication (5 Points)

SSH (Secure Shell) is a tool used to establish an encrypted connection to access and log into other devices remotely. In this task, you will set up an SSH server and learn the basic usage of SSH with key-based authentication.

Make sure to document all the steps and used commands.

- (a) Install an SSH server on one of the computers and an SSH client on the other. Create a user account on the server that you can use to log in from the client. Test your setup by logging in via SSH with a password. (1 Point)
- (b) To enable key-based authentication, generate an SSH key pair using the *ssh-keygen* tool on the client computer. Then, configure the server to allow the user to authenticate using the generated key. (2 Points)
- (c) Perform the following steps using key-based authentication (i.e. without entering a password):
 - 1) Use `scp` (secure copy) to transfer a text file from the client system to the server system.
 - 2) Rename the file on the server without logging in (remote command execution).
 - 3) Log in to the server system and edit the file.
 - 4) Use `scp` from the client to transfer the modified file back from the server to the client.

(2 Points)

Task 2 - SSH connection behind NAT via Relay Server (5 Points)

In the previous task, the setup only works when both computers have direct access to each other. However, if you're on the go and in a different network, you may encounter difficulties accessing the SSH server due to the Network Address Translation (NAT).

An approach to overcome this problem and remotely access your own computer is to bypass the NAT using SSH tunnels via an additional external relay server.

To accomplish this, you need to create an SSH tunnel between the computer running the SSH server and the relay server, as well as between the SSH client and the relay server. This enables the SSH client to establish an SSH connection to the SSH server through these tunnels.

Document all used commands and describe their effects.

- (a) Set up the SSH tunnel between the SSH server and relay server, as well as between the SSH client and the relay server. We will provide you with a relay server at *meet.dmi.unibas.ch*. There, you can log in with your university username and password. Next, configure the relay server to allow login using the previously generated key, eliminating the need to enter a password.

Note: Use *ssh -L/-R*. The man page¹ provides additional information on the *ssh* command. Make sure that the ports of the port mappings you specify are not already in use on the relay server. To avoid port conflicts, use the port number 51000 + the last three digits of your matriculation number. (E.g. If your matriculation number ends with 123, then use port 51123 for your port mapping.)

(2 Points)

- (b) Connect to the SSH server via the previously established SSH tunnels while you are in a different network than the server. Create a test file on the server remotely using SSH and transfer it to the client using *scp*. (1.5 Points)

- (c) Create a diagram showing how the IP address and port of a packet change on its way from the client, through the relay server, to the SSH server. (1.5 Points)

Task 3 - Bonus: Remote Desktop wit RealVNC (3 Bonus Points)

Nowadays, most applications are operated through a graphical user interface, so in many cases, a remote connection via the terminal is no longer sufficient. There are several applications available, such as RealVNC, that allow remote control of the graphical user interface. A client can connect to a RealVNC server and then remotely control it as if they were directly operating the computer on site. As there can also be issues with NAT when the server and client are in different networks, we use the same setup with the relay server as in Task 2.

- (a) Install RealVNC-Server² on the computer you want to control remotely and RealVNC-Viewer³ on the other computer. Connect to the machine running the RealVNC server using SSH tunneling (as in Task 2) and remotely control it while you are in a different network than the computer.

Note: To install the RealVNC-Server you are required to create an account. You don't need to provide your personal data and you can even use an e-mail address that does not exist. Otherwise you can also use another VNC Server/Client implementation.

(2 Points)

¹<https://linux.die.net/man/1/ssh>

²<https://www.realvnc.com/de/connect/download/vnc/>

³<https://www.realvnc.com/de/connect/download/viewer/>

- (b) If the connection between the relay server and the SSH server is briefly interrupted, it will cause the SSH tunneling to be interrupted and the SSH server will no longer be reachable from the outside. To avoid this scenario, you can use `autossh`⁴, which starts, monitors, and restarts SSH tunnels if they get interrupted, so that you can reconnect to the server.

Set up `autossh` for the computer running the RealVNC Server. (1 Point)

⁴<https://linux.die.net/man/1/autossh>